A Reproduced POYO Foundation Model: Transfer Limits and Data Augmentation as Implicit Spectral Regularization

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Abstract

POYO (Azabou et al., 2023) is a transformer foundation model for neural population decoding that tokenizes individual spikes and reads out behavior through per-unit and per-session embeddings. We report a faithful, end-to-end reproduction of the POYO-MP checkpoint—held-out test $R^2 = 0.904$ on the Perich–Miller center-out reaching data (Perich et al., 2018)—and then use the converged model as a base to ask two questions the original paper leaves open: where does its representation transfer, and why does training-time augmentation help. On transfer, three demonstrations confirm the foundation-model promise: with a frozen core and only new unit/session embeddings, the model decodes a held-out session from 10% of its data (R^2 0.72, vs. ≈ 0 from scratch), holds R^2 0.916 across 1114 days of chronic recordings with zero per-day retraining, and adapts to a completely unseen animal at R^2 0.881 (gap +0.89 over scratch). A fourth, deliberately adversarial test—transferring the motor-pretrained core to a cognitive interval-timing task (Sohn et al., 2019)—shows neutral-to-negative transfer (-0.013 R^2) that is robust to how the timing target is represented, mapping the edge of the model's competence. Pushed to the farthest domain—a frozen primate-motor core decoding rat entorhinal–hippocampal position across a species, region, and modality gap (Vollan et al., 2025)—transfer reappears, but strictly as a data-efficiency effect (session-mean +0.117 R^2, up to +0.311 on the most data-starved session and -0.012 where from-scratch already saturates): the frozen representation substitutes for training data wherever the target is expressible by the readout. Finally, we operationalize the linear-model theory of Lin et al. (2024)—data augmentation as implicit spectral regularization of the feature covariance—on this deep model. POYO's UnitDropout is exactly their "random-mask" augmentation; with a plastic feature extractor it spreads the readout-input covariance spectrum and cuts its condition number 2.6×, and this buys a +0.13 R^2 generalization gain precisely when training data is scarce, vanishing once the decode saturates. The effect lives in the feature covariance, not the weight spectrum—a label-free Heavy-Tailed Self-Regularization analysis of the weights barely moves—scoping where the two spectral views agree. We release the full reproducible-document pipeline (data → figures → manuscript).

1 Introduction

Foundation models for neural data promise a single set of weights that decodes behavior across sessions, subjects, and tasks. POYO (Azabou et al., 2023) is a leading instance: it represents a recording as an unordered set of spike tokens, attends them into a fixed bank of latent queries, and reads out continuous behavior, with InfiniteVocabEmbedding tables that assign every recorded unit and every session its own learned vector. This design lets the same transformer ingest arrays with different channel counts and lets a new recording be "identified" into the model by learning only its embeddings (the §2.5 unit-identification protocol of Azabou et al., 2023), freezing the transformer core.

Reproducing such a model is a precondition for studying it, and a non-trivial one: our own earlier attempts repeatedly produced weight collapse (most transformer matrices decaying to zero during training) and R^2 near zero. We show this was an environment artifact, not a property of the recipe, and obtain a converged checkpoint matching the paper. We then treat the converged model as an object of study and contribute:

- A faithful reproduction (Section 2): the unmodified nerdslab/poyo repository, in its pinned environment, on 8×H100 with the exact recipe (global batch 1024, max-LR 0.032, fp32, SparseLamb), reaches R^2 0.904.
- Three transfer demonstrations (Section 3) that quantify the foundation-model payoff—data efficiency, chronic stability over 1114 days, and cross-subject transfer to an unseen animal—one adversarial test (Section 4) that finds its limit on cognitive timing, and a farthest-domain cross-species test (Section 5, monkey motor → rat spatial coding) that together show the frozen core's payoff is a data-efficiency effect gated on whether the target is expressible by the readout.
- An empirical bridge (Section 6) from the linear data-augmentation theory of Lin et al. (2024) to a deep neural foundation model: POYO's UnitDropout regularizes the feature-covariance spectrum and buys generalization under data scarcity, with the effect localized to representations rather than weights.

Throughout we cross-reference a label-free weight-spectrum diagnostic, wwj, our Bayesian reimplementation of WeightWatcher-style Heavy-Tailed Self-Regularization (Martin & Mahoney, 2021; Martin et al., 2021), to ask whether the augmentation signature is visible in the weights as well as the features.

2 Faithful reproduction of POYO-MP

The collapse was the environment, not the recipe. Across many earlier runs, most weight matrices decayed to (numerically) zero during training and decode R^2 stalled near zero. We ruled out weight-decay-on-embeddings, distributed training, learning-rate/batch choices, and precision (single-GPU fp32 collapsed too); our configs were byte-identical to the authors'. The residual difference was the software environment—a newer PyTorch under which the custom SparseLamb optimizer's masked trust-ratio update behaves differently. Running the unmodified repository in its pinned environment (torch∼2.10, brainsets@4ccee58) on a dedicated, non-preemptible 8×H100 node, in a single uninterrupted pass of the exact recipe, removed the collapse on the first try.

Result. The converged POYO-MP (11.8M parameters, 1000 epochs) reaches held-out test $R^2 = 0.904$ on Perich–Miller center-out reaching, matching the published model. As an independent, label-free check we ran wwj over the trained weights: mean Heavy-Tailed power-law exponent $\bar{\alpha} = 1.62$ with 17% of fittable layers inside the $\alpha = 2$ robustness region, versus an undertrained checkpoint's far-from-2 spectrum—the HTSR training-quality signature, confirmed on a real foundation model. Of 106 weight matrices, 98 are spectrally degenerate (rank-deficient), reflecting POYO's inherently low-rank attention blocks; the decode flows through the survivors, the residual stream, and the embeddings.

3 What transfers: three foundation-model demonstrations

We finetune the converged base to new data using the §2.5 protocol—load the pretrained weights, extend_vocab for new units/sessions, freeze the transformer core, and train only the embeddings (and, where the readout is new, the readout head). The contrast in every demonstration is the same architecture trained from scratch on the same new data.

Data efficiency. On a held-out area2_bump session (Chowdhury et al., 2020; Pei et al., 2021), varying only the fraction of new-session training data (matched LR/batch across arms), the pretrained model decodes hand velocity at R^2 0.72 from just 10% of the session, rising to 0.85 at full data; the from-scratch model never exceeds $|R^2| < 0.001$ at any fraction

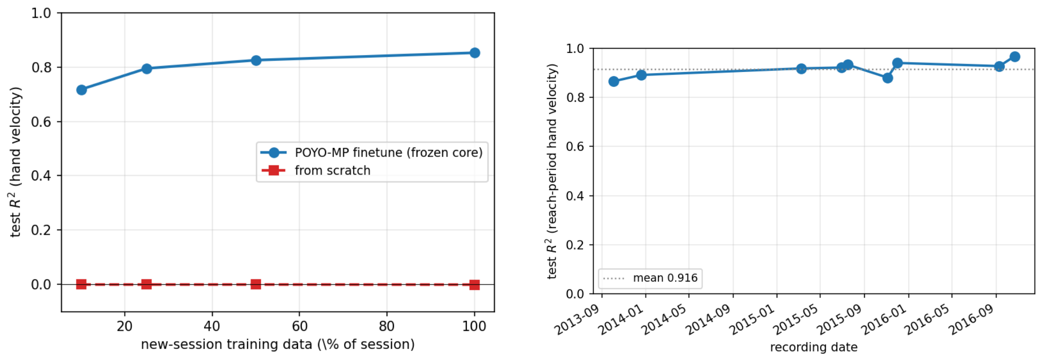

Figure 1: Left: data efficiency. Pretrained POYO-MP (frozen core) vs. from scratch as a function of new-session training data on a held-out area2_bump session. Right: chronic stability. One frozen base, zero retraining, across 1114 days of Chewie recordings.

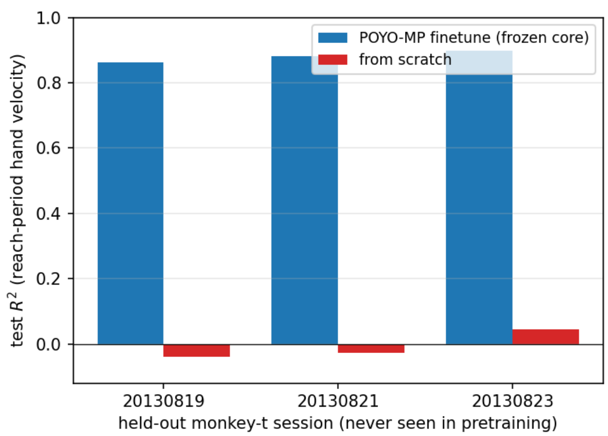

Figure 2: Cross-subject transfer to held-out monkey t (never seen in pretraining). Finetuning only the embeddings of the frozen base recovers ∼0.88 R^2; the same architecture from scratch on one session does not learn.

(Figure 1, left). A single small session is simply too little data to train an 11.8M-parameter transformer; the foundation model converts it into a few embedding vectors.

Chronic stability. Freezing the base entirely (zero per-day retraining) and evaluating it on 9 Chewie center-out sessions spanning 2013–2016, decode R^2 holds at mean 0.916 (range 0.865–0.967) across 1114 days (Figure 1, right). One set of weights sustains ∼0.9 R^2 over three years of electrode drift and unit turnover, absorbed by the per-unit embeddings.

Cross-subject transfer. The base was trained on 99 sessions from three monkeys; monkey t was entirely held out. Finetuning embeddings on a single unseen-animal session decodes at mean R^2 0.881 (range 0.863–0.899 over 3 sessions), versus −0.01 from scratch—a gap of +0.89 (Figure 2). The model adapts to a new animal it has never seen at ∼0.88 R^2, while the same architecture trained on one session learns nothing.

Transfer and the weight spectrum agree. Table 1 pairs the decode R^2 with the wwj weight-spectrum verdict on the same checkpoints. The frozen-core finetune inherits the base spectrum identically ($\bar{\alpha} = 1.62$, 17% in-ROPE); the from-scratch model shows an undertrained heavy-tail ($\bar{\alpha} = 3.31$, 0% in-ROPE) and decodes poorly. The label-free spectral quality and the decode quality move together: motor→motor transfer is a large, coherent win.

Table 1: Two-axis transfer summary (motor→motor, area2_bump). Decode R^2 and the wwj weight-spectrum verdict are read off the same checkpoints. ROPE = region of practical equivalence around the $\alpha = 2$ robustness fixed point.

Model	decode R^2	$\bar{\alpha}$	in-ROPE	degen/106
Base (perich, 99 sessions)	0.904	1.62	17%	98
area2 finetune (frozen core)	0.834	1.62	17%	98
area2 from scratch	0.313	3.31	0%	0

4 Where it breaks: motor-to-timing transfer

A foundation model is as informative in its failures as its successes. POYO is, at heart, spike tokenization plus a per-timestep continuous readout; cognitive interval timing, by contrast, lives in latent dynamics—ramp-to-threshold trajectories whose speed scales with the produced interval (Sohn et al., 2019; Merchant et al., 2013). We therefore transfer the motor-pretrained core to the dmfc_rsg Ready-Set-Go task, decoding a time-to-go ramp (the dynamics-engaging target: a descending ramp from the sample interval at Set to zero at Go).

The motor-pretrained frozen core gives no advantage: finetune R^2 0.887 vs. from-scratch 0.900 (-0.013). The result is robust to target representation—a static constant-interval target shows the same ordering (0.798 vs. 0.845). The sign of transfer thus flips with task distance: motor→motor is +0.89 to +0.52, whereas motor→cognitive-timing is neutral-to-negative. Notably the wwj ↔decode relationship inverts here too: the from-scratch timing model sits at a healthier-looking $\bar{\alpha} = 2.63$ yet decodes better than the frozen-core finetune ($\bar{\alpha} = 1.62$ inherited from the motor base)—so α reflects the training regime, and the "healthy-α →good-decode" correlation is domain-match-mediated, not universal. POYO's value is cross-session and cross-subject alignment of motor representations, not the timing computation itself.

5 The farthest domain: cross-species spatial transfer

Timing fails because the target is not expressible by a per-timestep readout, not because the domain is far. To separate these two explanations we push to the farthest point on the transfer axis that still has a readout-expressible target—a different species, brain region, and behavior at once. We transfer the monkey motor-cortex base to rat entorhinal–hippocampal recordings (Vollan et al., 2025) (Neuropixels in MEC and hippocampus during open-field foraging) and decode 2D allocentric position—the grid/place-cell readout—through a new position_2d head, on 3 held-out sessions, again contrasting frozen-core finetune against from scratch.

Unlike timing, rat position is decodable from spike tokens: from scratch reaches R^2 0.777 on average, up to 0.947. And the frozen monkey-motor core does transfer across the species gap—finetune R^2 0.819–0.935 (mean 0.893): a frozen primate-motor backbone with only a fresh position head reads out rat spatial coding. But the benefit is not uniform (Figure 3, Table 2); it tracks the from-scratch deficit. On the data-starved session (496 train windows, where scratch reaches only R^2 0.508) the pretrained core lifts decode by +0.311; where scratch already saturates (1858 windows, R^2 0.947) it adds nothing (-0.012). The session-mean gain is a modest +0.117, but its distribution is the point: POYO-MP's frozen representation buys decode accuracy in proportion to how starved the from-scratch model is.

This reconciles the whole transfer-vs-distance axis. The frozen core's payoff is a data-efficiency effect—it substitutes for training data—wherever the target is expressible by POYO's per-timestep readout, from motor velocity (Section 3) to rat position here; it is largest under scarcity (matching both the area2 data-efficiency curve and the augmentation×data-fraction interaction of Section 6), and it is absent on cognitive timing (Section 4) for a different reason: there the target itself eludes the readout, so neither

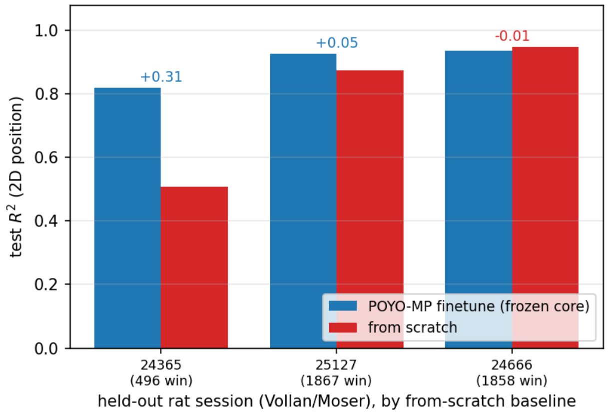

Figure 3: Cross-species spatial transfer (monkey motor → rat MEC/hippocampus), decoding 2D position through a new head, on 3 held-out rat sessions ordered by from-scratch baseline. Frozen-core finetune (blue) vs. from scratch (red), with Δ annotated. The transfer gain is large on the data-starved session (fewest train windows) and vanishes where from-scratch already saturates—a data-efficiency effect, not a uniform cross-species boost.

Table 2: Cross-species spatial transfer (Vollan/Moser rat, 2D position via a new position_2d head), sorted by from-scratch baseline. The frozen-core finetune gain Δ tracks the from-scratch deficit: largest where data is scarce (fewest train windows), negligible where scratch saturates.

Rat session	train windows	scratch R^2	finetune R^2	Δ
of_24365_2	496	0.508	0.819	+0.311
of_25127_1	1867	0.875	0.926	+0.051
of_24666_1	1858	0.947	0.935	-0.012

transfer nor scratch succeeds. Distance per se does not set the sign of transfer; readout-expressibility of the target and from-scratch data scarcity do.

6 Data augmentation as implicit spectral regularization

Lin et al. (2024) prove, for linear models, that data augmentation acts as implicit spectral regularization of the data/feature covariance: it reweights the eigenvalue proportions in a data-dependent way and uniformly boosts the spectrum (a ridge-like effect). Their canonical "random-mask" augmentation is exactly POYO's UnitDropout, which drops a random subset of units per sample. We ask whether their linear-model prediction survives in a deep neural foundation model, instrumenting the input to POYO's linear readout—the natural "feature" whose covariance the theory concerns—and measuring its spectrum across three UnitDropout strengths (off / standard / aggressive).

A plastic extractor is required (the null control). With the core frozen (the §2.5 protocol), augmentation cannot reshape the feature covariance, because the features are produced by fixed weights: the effective rank moves by 0.024 across all three levels and the wwj weight-$\bar{\alpha}$ is identical (1.78) by construction. This is a clean negative control—and a sanity check that freeze-core truly freezes.

With a plastic extractor, the theory reproduces. Finetuning all weights, the predicted signature appears (Table 3, Figure 4): as UnitDropout strengthens, the feature-covariance effective rank rises (1.96 → 2.02), the top eigenvalue de-concentrates (0.63 → 0.56), and the condition number falls monotonically from 1.1×10^{12} to 4.3×10^{11}—a 2.6× reduction.

Table 3: Finetune-all UnitDropout ablation (held-out monkey t). As augmentation strengthens, the readout-input feature covariance spreads (effective rank up, top-1 fraction down, condition number down 2.6×), while the wwj weight-spectrum $\bar{\alpha}$ is nearly constant. Decode R^2 is flat because the full-data decode is near-saturated.

UnitDropout	test R^2	eff. rank	top-1 frac	cond. number	wwj $\bar{\alpha}$
off	0.856	1.96	0.63	1.1×10^{12}	1.770
standard	0.851	2.02	0.55	8.8×10^{11}	1.767
aggressive	0.865	2.02	0.56	4.3×10^{11}	1.762

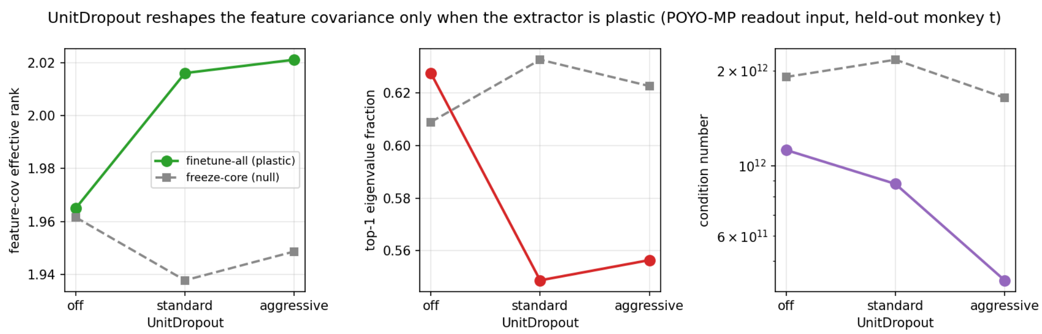

Figure 4: UnitDropout reshapes the feature covariance only when the extractor is plastic. Finetune-all (solid) spreads the spectrum and cuts the condition number; freeze-core (dashed) is flat—the null control. The effect is the Lin et al. (2024) spectral-regularization prediction realized in POYO's representations.

This is the Lin et al. "eigenvalue-proportion reweighting" effect, reproduced on a real deep model.

The signature is in the features, not the weights. Strikingly, the wwj weight-spectrum exponent barely moves over the same runs (1.770 → 1.762). At this finetuning budget the augmentation's fingerprint is written into the representation covariance—the object the theory is about—while the weight spectrum changes far more slowly. The two spectral views (feature covariance vs. weight HTSR) are therefore not equivalent; this scopes precisely where a weight-only diagnostic like wwj will and will not see an augmentation effect.

The spectral spreading buys generalization—when data is scarce. A spread spectrum should help most when the decode is not saturated. We close the loop with a {data fraction} × {UnitDropout} grid (Table 4, Figure 5). At 10% of the session, augmentation lifts test R^2 from 0.539 to 0.669 (monotone, +0.13), and the off-augmentation model is the most ill-conditioned of the entire grid (3.1×10^{12}), which augmentation cuts ∼3.7× (to 8.2×10^{11})—the conditioning improvement tracks the generalization gain. The gap closes by 25% data and is flat-to-slightly-negative at full data (0.868 vs. 0.863). This is the textbook data×augmentation interaction, now demonstrated end-to-end on a neural foundation model: UnitDropout (= the Lin et al. random-mask augmentation) → feature-covariance spectral regularization → a generalization gain, gated on both a plastic feature extractor and data scarcity.

7 Discussion

A faithful reproduction turned POYO-MP from a target into an instrument. Its representation transfers strongly across sessions, time, and subjects within the motor domain—a few embedding vectors recover ∼0.9 R^2 from 10% of a session, hold it for three years, and port to an unseen animal—and the label-free weight-spectrum quality tracks decode quality in that regime. The same representation does not confer an advantage on cognitive interval

Table 4: Low-data generalization grid (finetune-all, held-out monkey t, fixed batch size): test R^2 vs. UnitDropout strength at three training-data fractions. The augmentation benefit is large under scarcity and vanishes as the decode saturates.

train data	off	standard	aggressive	Δ(agg−off)
10%	0.539	0.604	0.669	+0.130
25%	0.777	0.776	0.784	+0.007
100%	0.868	0.841	0.863	-0.005

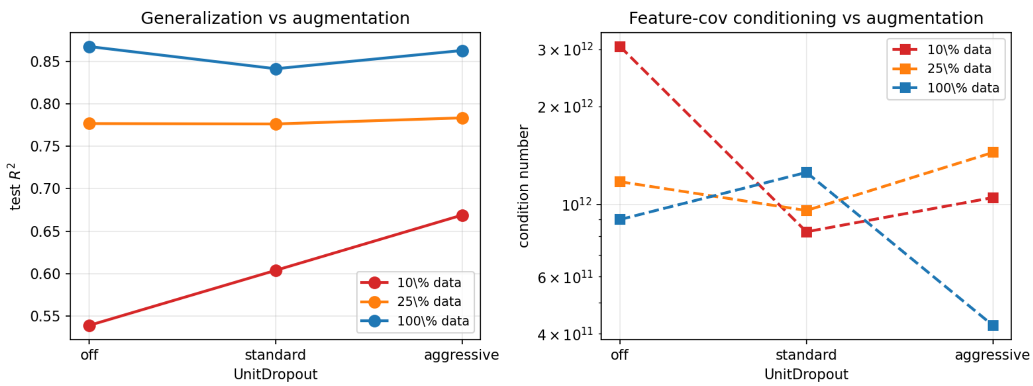

Figure 5: Augmentation buys generalization exactly when data is scarce. Left: test R^2 vs. UnitDropout, one line per training-data fraction. Right: the feature-covariance condition number—most ill-conditioned at 10% data with no augmentation, and cut sharply by UnitDropout, mirroring the generalization gain.

timing, regardless of how the timing target is framed, marking a concrete boundary of its competence and a case where the spectral↔decode correspondence inverts. Pushed across a full species gap to rat spatial coding (Section 5), the core transfers again—but strictly as a data-efficiency effect, largest where the from-scratch model is starved—so what gates transfer is not domain distance but whether the target is expressible by POYO's readout and how scarce the new data is.

The augmentation study connects a clean piece of linear theory (Lin et al., 2024) to a working deep model and pins down its boundary conditions. The predicted spectral regularization of the feature covariance is real and the generalization payoff is real, but both are gated: the feature extractor must be plastic, and the decode must not be saturated. The most useful caveat for practitioners of weight-spectrum diagnostics is the asymmetry we observe—at a realistic finetuning budget, augmentation's signature is in the representation covariance, not the weight spectrum—so the feature-covariance and weight-HTSR views answer different questions and should not be conflated. The main limitations are scope: the cross-subject and augmentation results use single held-out sessions of one animal, and the weight-spectrum measurements are at a finetuning (not from-scratch) budget; a multi-animal held-out test set and a from-scratch augmentation sweep would sharpen the generality of both claims.

Reproducibility. Every number, table, and figure in this paper is produced from the raw experiment logs by a single tangle script (paper/generate.py); build.sh regenerates the manuscript end-to-end. Nothing is hand-transcribed.

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